Practice Problems

1. Learn some use-case of INT_MAX and INT_MIN. Although it was not explicitly discussed in the lecture class yet, but you may have needed it for solving some problems in the lab.
2. Practice various shapes to print using nested loops. If you can understand the working principles of these and can implement it by yourself (not by memorizing, but understanding how it actually works), then hopefully you can solve any patterns for the exam. Some examples: (Not limited to these only. These are just samples).
   i.
```
    A
   ABA
  ABCBA
 ABCDCBA
ABCDEDCBA
```
   ii.
```
1 1 1 1 1
1 0 0 0 1
1 0 0 0 1
1 0 0 0 1
1 1 1 1 1
```
   iii.
```
ABCDEDCBA
ABCD DCBA
ABC   CBA
AB     BA
A       A
```
   iv. You can find more examples (some with solutions) in this link: https://www.ankitweblogic.com/c/exerciseonnestedloop.php
   v. You can practice previous mid-final-quiz questions too.
   vi. You may practice problems from regular lab tasks
3. Find the sum of the digits of a number
4. Count the number of digits of a number
5. Count the number of odd and even digits of a number
6. Print binary of a positive integer value
7. Find whether a given number is palindrome or not
8. Find whether a given number is prime or not
9. Find whether a number is perfect number or not
10. Print an integer array in reverse order
11. Print only odd numbers from an array
12. Print numbers from an array that are more than 10
13. Find the sum of odd and even numbers of an array and find the difference between the sum
14. Find the index of vowels in a character array
15. Print the characters from the odd index from a character array
16. Add the values of 2 arrays from same index together and save into a 3rd array
17. Find the difference between corresponding index of 2 integer arrays and save the values into 3rd array

18. If the sum of corresponding location of 2 integer arrays exceeds the value of 10, then store 1, otherwise store 0 into the same index of 3rd array
19. Count the number of appearances of a number into an integer array, and find the indices of its appearances.
20. Increment the values of an integer array by 10, if the number is divided by 3 and 5
    Find the number of prime numbers in an array

NB: In exam, the question may not be written as direct as it is written here. Usually, the problems come with a description or a scenario. You have to figure out what category that problem belongs to and how to solve it.
Please remember, none of the problems given above may appear in the exam. But if you practice these, it will definitely help a lot.